3GPP TSG RAN WG1 Meeting #99 **R1-1913224**
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Reno, NEVADA, November 18th – 22nd, 2019

Source: **ESA**
Title: **NTN system level and link budget calibration: DL and UL results**
Agenda Item: **7.2.5.1**
Document for: **Discussion and Approval**

1 Introduction

The SI on "Study on Solutions for NR to Support Non-Terrestrial Networks" was approved in RAN#80 meeting [1] and further updated in RAN#82 meeting [2]. The RAN1 objectives have been reported below.

- *Consolidation of potential impacts as initially identified in TR 38.811 and identification of related solutions if needed [RAN1]:*
    - *Physical layer control procedures (e.g. CSI feedback, power control)*
    - *Uplink Timing advance/RACH procedure including PRACH sequence/format/message*
    - *Making retransmission mechanisms at the physical layer more delay-tolerant as appropriate. This may also include capability to deactivate the HARQ mechanisms.*
- *Performance assessment of NR in selected deployment scenarios (LEO based satellite access, GEO based satellite access) through link level (Radio link) and system level (cell) simulations [RAN1]*

In RAN1 #98 meeting, several parameters (from Table 6.1.1-1 to Table 6.1.1-6 in [3]) have been agreed for calibration in NTN performance evaluation. Further clarifications have been also added after RAN1 #98bis meeting [8].

Essentially, this contribution represents the new version of the document R1-1911254 [6], where all numerical results for downlink and uplink system level calibration (**Section 3**) have been collected (please refer to Table 6.1.1-9 in [3] for the agreed satellite scenarios).

In addition, this contribution shows also in **Section 4** the results related to the link-budget calibration, as described in Sec. 6.1.3 of [3] and taking into account the RAN1 #98bis agreements [8].

2 System Level Configuration and Assumptions

In this document, the system-level configuration is based on the agreed parameters and methodology described in [3] and taking into account the latest RAN1 #98bis agreements [8].

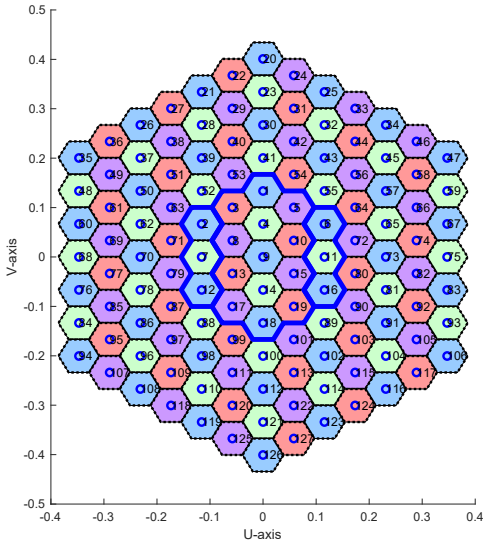

Figure 1 - Beam layout for LEO at 600 km systems with central beam at nadir in the UV-plane and Frequency Reuse Option 2. The blue line identifies the 19 beams retained for the numerical assessment. [Case 7]

As for the configuration of the beam layout across the coverage area, 61 beams were considered for FFR and 127 for frequency reuse options 2 and 3; these configurations correspond to having 4 and 6 tiers of beams, respectively. The beam centers are spaced according to the Adjacent Beam Spacing (ABS); the numerical results and the related statistics were gathered from the 19 central beams only. Figure 1 shows an example of 127-beams deployment, for Frequency Reuse (FR) option 2 (*i.e.*, 3-colour FR, or FR3), in which the 19 central beams retained for numerical assessments are enclosed in the thick blue line. The beam layout is defined on the UV-plane, [3], and then projected in (LAT, LON) coordinates on the spherical Earth's surface.

2.1 Downlink computation

With respect to the user deployment, it shall be noticed that, in the downlink, the interference towards each UE is generated by the interfering transmitting antennas on the satellite; thus, interference is defined by geometry, *i.e.*, it is fixed given the receiving UE location and the beam layout configuration, since it depends on the angle between the UE location and the interfering antenna boresight.

This is highlighted in Figure 2, where:

i. $\vartheta_{b,u}^{(sat,b)}$ denotes the angle between the u-th user in the b-th beam and the intended b-th antenna boresight;

ii. $\vartheta_{b,u}^{(sat,i)}$ denotes the angle between the u-th user in the b-th beam and the i-th interfering antenna boresight.

Consequently, the UEs are generated on a (LAT, LON) grid and 100 Monte Carlo iterations were considered, in order to average on stochastic effects (*e.g.*, shadowing). In particular, depending on the satellite orbit and altitude, the following grid-point resolution has been considered: i) GEO, 0.05 degrees (*i.e.*, 9750 UEs in the beam layout); ii) LEO at 600 km, 0.01 degrees (*i.e.*, 3400 UEs); and finally iii) LEO at 1200 km, 0.02 degrees (*i.e.*, 3400 UEs). At each Monte Carlo iteration, the link budget is computed in all grid points.

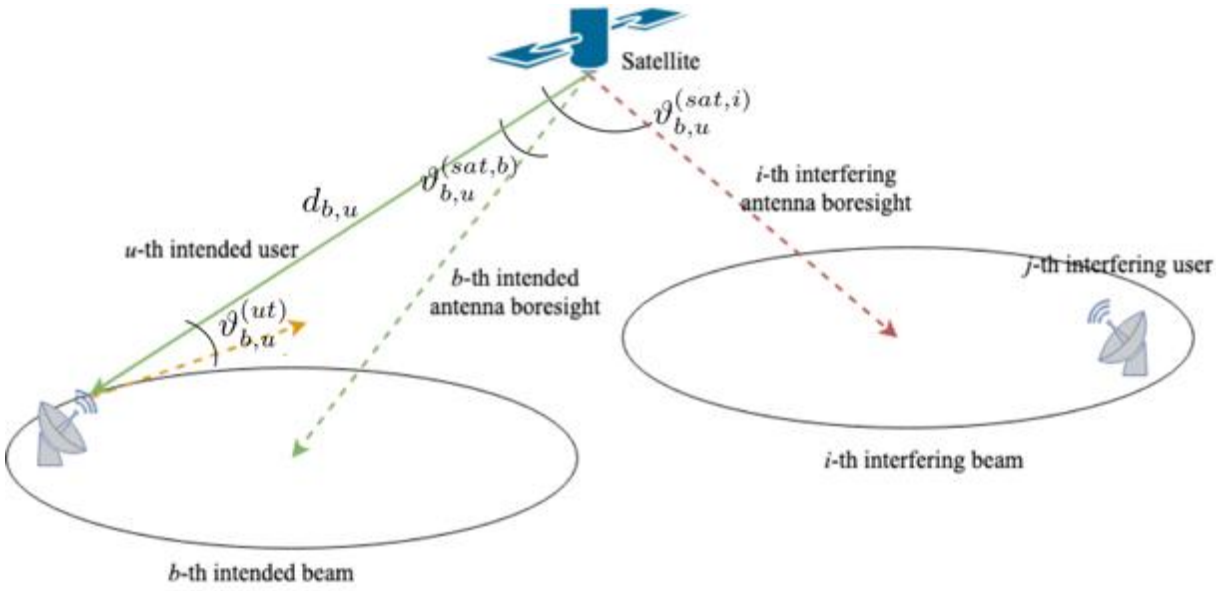

Figure 2 - Geometry for downlink link budget computation.

Based on the above considerations, the link budget is computed as follows (see Section 6.1.3.1 in [3]):

$$CNR[dB] = EIRP[dBW] + \frac{G}{T}[dB] - k[dBW/K/Hz] - PL_{FS}[dB] - PL_A[dB] - PL_S[dB] - PL_{AD}[dB] - 10\log_{10}\left(\frac{BW[Hz]}{K_{FR}}\right)$$

where K_{FR} is the bandwidth factor based on the selected frequency reuse scheme and

$$EIRP[dBW] = EIRP_{den}[dBW/MHz] + RP^{(sat)}\left(\vartheta_{b,u}^{(sat,b)}, a_{sat}, f\right) + 10\log_{10}\left(\frac{BW[MHz]}{K_{FR}}\right)$$

$$\frac{G}{T}[dB] = G_{RX}[dB] + RP^{(UT)}\left(\vartheta_{b,u}^{(sat,b)}\right) - N_f[dB] - 10\log_{10}\left(T_0[K] + (T_a[K] - T_0[K])10^{-0.1N_f[dB]}\right)$$

with: i) $RP^{(sat)}\left(\vartheta_{b,u}^{(sat,b)}, a_{sat}, f\right)$ being the normalised transmitting radiation pattern in [dB], *i.e.*, always less or equal to 0 dB, from the intended antenna towards the u-th UE introduced to take into account the fact that the UE might not be at the beam antenna boresight; ii) $RP^{(UT)}\left(\vartheta_{b,u}^{(sat,b)}\right)$ the normalised receiving radiation pattern in [dB], which depends on the UE antenna type; and iii) G_{RX} the maximum receiving antenna gain.

In all of the equations, it is considered, as per TR 38.821 [3], that the UE antennas have ideal pointing; thus, the receiving antenna radiation pattern, $RP^{(UT)}\left(\vartheta_{b,u}^{(sat,b)}\right)$, is always equal to 0 dB.

For the sake of completeness, Figure 3 shows the radiation pattern for VSAT; the handheld antenna type is omnidirectional, so it is not shown in the following. It shall also be noticed that for handheld terminals it is assumed that a combination of the two Rx branches allows to prevent depolarization loss due to the usage of linear polarized antennas.

Observation 1: Handheld have been assumed with two RX-branches, therefore no polarization loss in the downlink results.

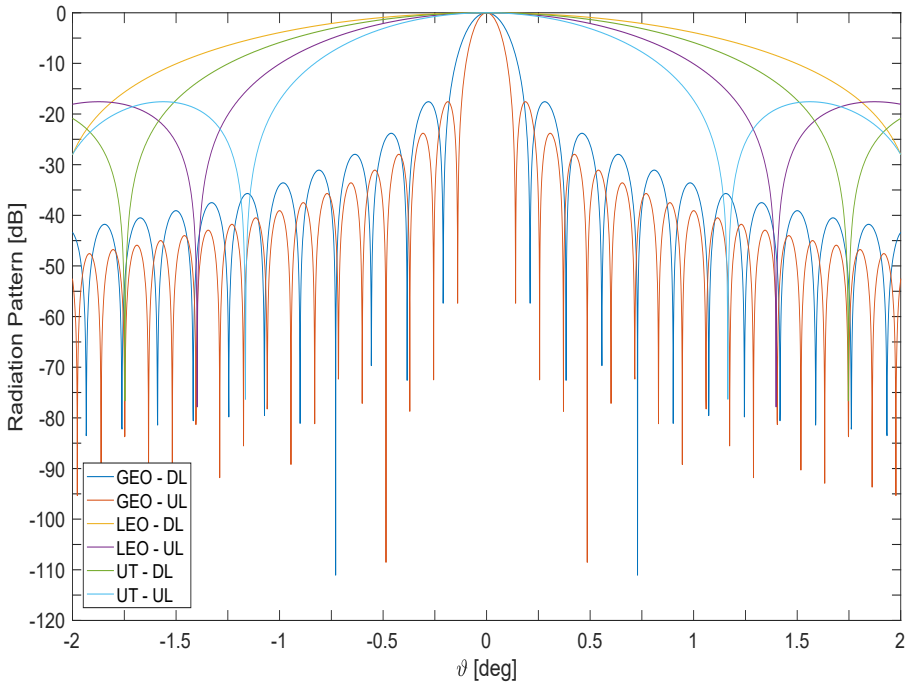

Figure 3 - VSAT UE type: receiving antenna radiation pattern, [7].

The overall path loss consists of the following terms:

- PL_{FS}: free space loss.
- PL_A: atmospheric gas losses, computed based on ITU-R P.676 Annex 2.
- PL_S: shadowing, computed as reported in [7], Section 6.6.2, and assuming 100% LOS probability.
- PL_{AD}: additional losses assumed to be due to scintillation phenomena, as per Section 6.6.6. in [7]. For LEO systems in S-band, it has been assumed that all users are located between 20 and 60 degrees latitude, which leads to a null ionospheric scintillation term. On the other hand, the tropospheric scintillation loss (Ka-band) is computed from the attenuation level at 99% of the time presented in Table 6.6.6.2.1-1 of [7].

With respect to the interfering terms in the CIR and CINR, the interference from the generic *i*-th antenna towards the *u*-th user is computed as:

$$I_i[dB] = EIRP_i[dBW] + G_{RX}[dB] - PL_{FS}[dB] - PL_A[dB] - PL_S[dB] - PL_{AD}[dB]$$

where:

$$EIRP_i[dBW] = EIRP_{den}[dBW/MHz] + RP^{(sat)}\left(\vartheta_{b,u}^{(sat,i)}, a_{sat}, f\right) + 10\log_{10}\left(\frac{BW}{K_{FR}}\right)$$

with $RP^{(sat)}\left(\vartheta_{b,u}^{(sat,i)}, a_{sat}, f\right)$ being the normalised radiation pattern in [dB], i.e., always less or equal to 0 dB, from the *i*-th interfering antenna towards the *u*-th UE. These terms are then combined together, yielding to the following CIR and CINR equations:

$$I[dB] = 10\log_{10}\left(\sum_{i=1}^{N_{int}} 10^{0.1 I_i[dB]}\right)$$

$$CIR[dB] = C[dB] - I[dB]$$

$$CINR[dB] = -10\log_{10}\left(10^{-0.1CNR[dB]} + 10^{-0.1CIR[dB]}\right)$$

where N_{int} is the number of interfering beams, which depends on the considered FR option. Finally, the coupling loss (CL) is computed as:

$$Closs[dB] = PL_{FS}[dB] + PL_A[dB] + PL_S[dB] + PL_{AD}[dB] - \left(RP^{(sat)}\left(\vartheta_{b,u}^{(sat,b)}, a_{sat}, f\right) + G_{TX}[dB]\right) - G_{RX}[dB]$$

In all of the above equations, it is considered, as per TR 38.821 [3], that the UE antennas have ideal pointing. Thus, the receiving antenna radiation pattern is always equal to 0 dB.

2.2 Uplink computation

In this case, interference is not defined by a simple geometry, due to the fact that the interfering signals are received at the satellite and, thus, the combined effect depends on the specific location of the active UEs. In our software implementation, the following assumptions hold:

i) the UEs are deployed in the same grid as that defined above for the downlink;

ii) at each drop, a single user per beam is randomly selected for transmission, *i.e.*, a random scheduling is taken into account;

iii) the assigned bandwidth of each user in the uplink is 1/10 of the overall selected user bandwidth, so as to take into account that, in each beam, 10 UEs are present, as per [3]. By iterating over a sufficiently large number of drops, the results are equivalent to the case in which 10 UEs per beam are randomly deployed and then selected for transmission.

iv) 50000 Monte Carlo simulations are considered for generating a sufficient and stable uplink statistics.

The geometry for the uplink link budget calibration is shown in Figure 4. While the UEs are always assumed to be ideally pointing towards the satellite, *i.e.*, $\vartheta_{i,j}^{(ut)} = \vartheta_{b,u}^{(ut)} = 0°$, with $\vartheta_{i,j}^{(ut)}$ and $\vartheta_{b,u}^{(ut)}$ being, respectively, the angles between the antenna pointing of the *i*-th user in the *j*-th interfering beam and the satellite direction, and that for the intended *u*-th user in the *b*-th beam.

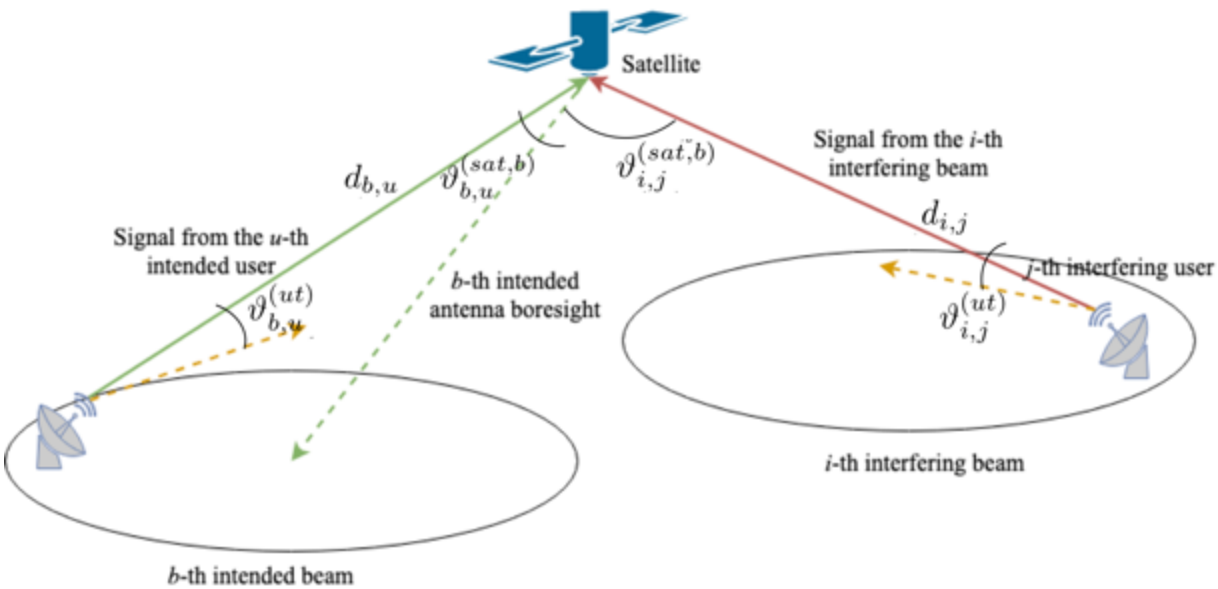

Figure 4 - Geometry for uplink link budget computation.

The computation of the CNR, CIR, and CINR is performed similarly to that reported in the downlink case. In particular, from Section 6.1.3.1 in [3]:

$$CNR[dB] = EIRP[dBW] + \frac{G}{T}[dB] - RP^{(sat)}\left(\vartheta_{b,u}^{(sat,b)}, a_{sat}, f\right) - k[dBW/K/Hz] - PL_{FS}[dB] - PL_A[dB] - PL_S[dB] - PL_{AD}[dB] - 10\log_{10}\left(\frac{BW[Hz]/10}{K_{FR}}\right)$$

where K_{FR} is the bandwidth factor based on the selected frequency reuse scheme, and

$$EIRP[dBW] = P_{TX} + G_{TX}$$

with P_{TX} and G_{TX} being the UE transmission power and maximum transmission antenna gain, respectively.

The value for $\frac{G}{T}[dB]$ is directly provided in [3]. As also highlighted previously, in all of the equations, it is assumed that the UE antennas have ideal pointing and, thus, the related UE radiation patterns are always equal to 0 dB. The path loss is computed as for the downlink.

With respect to the interfering terms in the CIR and CINR, the interference from the generic *i*-th interfering user in the *j*-th beam on the intended *b*-th signal at the satellite is computed as:

$$I_{b,u,i}[dB] = EIRP[dBW] + G_{RX}\left(\vartheta_{i,j}^{(sat,b)}\right) - PL_{FS,i,j} - PL_{A,i,j}[dB] - PL_{S,i,j}[dB] - PL_{AD}[dB]$$

where the path loss components and the satellite receiving antenna gain now depend on the specific interfering UE, while the EIRP is the same for all UEs as discussed above. These terms are then combined together, yielding to the same CIR, CINR, and Coupling Loss equations as in the downlink case.

Finally, for clarity, in case of handheld terminals, the polarization loss has been assumed following the agreement in [8]:

- 3dB depolarization loss should be taken into account assuming polarization reuse is applied and satellite reception implements circular polarization (e.g., for frequency reuse 4 case)
- 0dB depolarization loss can be assumed when satellite reception implements dual polarization per beam (e.g., for the frequency reuse 1 and 3 cases)

Observation 2: For uplink results with handheld terminals, 3dB polarization loss has been considered only in case of frequency reuse 4.

3 System Level Simulation Results for Calibration

In the following, the CNR / CIR / CINR / CL cumulative distribution functions are reported for Set-1. All data values (both Set-1 and Set-2) have been collected in the attached MS Excel file.

In particular, as far as the frequency reuse options are concerned, the following notation is adopted in all plot legends: full FR (FFR) is *option-1*, FR 3 (FR3) is *option-2*, and FR 4 (FR4) is *option-3*. In addition, the following bandwidth assumptions are considered:

- DL bandwidth: $30/K_{FR}$ MHz (in S band), $400/K_{FR}$ MHz (in Ka-band)
- UL bandwidth: 360 kHz (in S band), $400/10/K_{FR}$ MHz (in Ka-band)

3.1 GEO

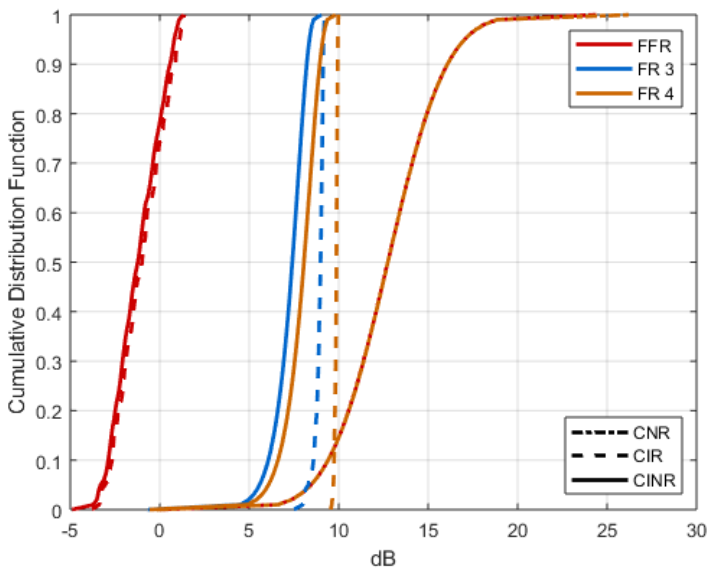

Figure 5 - GEO downlink link budget calibration. Setup: target elevation angle at 45 degrees, VSAT terminals, Ka-band. [Cases 1, 2, 3].

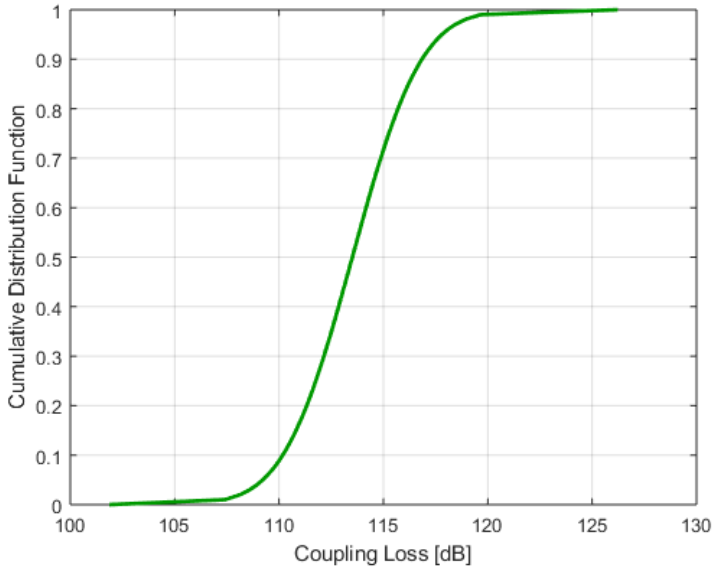

Figure 6 - GEO downlink coupling loss. Setup: target elevation angle at 45 degrees, VSAT terminals, Ka-band. [Cases 1, 2, 3].

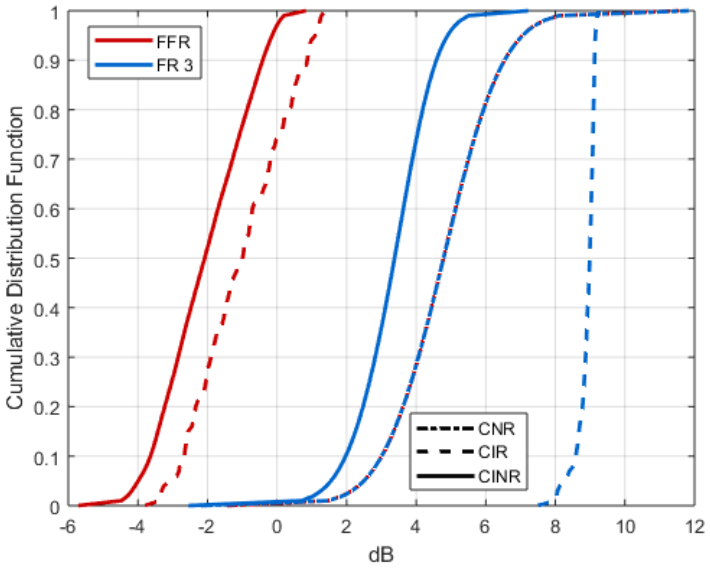

Figure 7 - GEO downlink link budget calibration. Setup: target elevation angle at 45 degrees, handheld terminals, S-band. [Cases 4, 5].

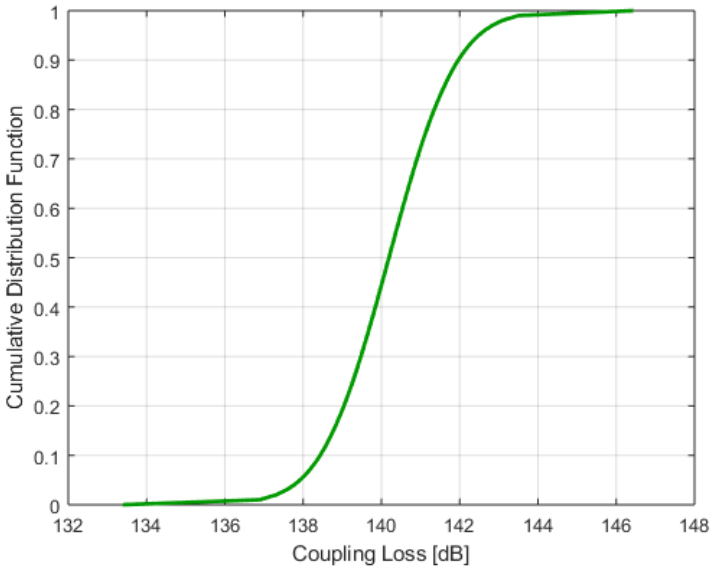

Figure 8 - GEO downlink coupling loss. Setup: target elevation angle at 45 degrees, handheld terminals, S-band. [Cases 4, 5].

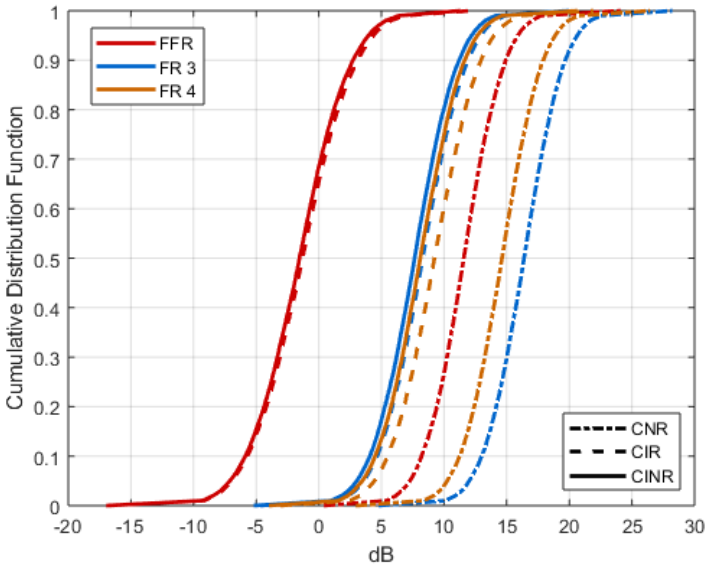

Figure 9 - GEO uplink link budget calibration. Setup: target elevation angle at 45 degrees, VSAT terminals, Ka-band. [Cases 1, 2, 3]

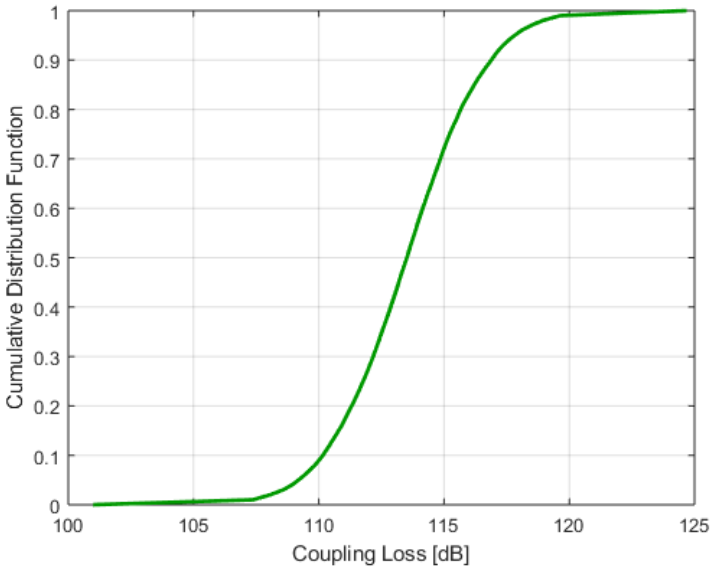

Figure 10 - GEO uplink coupling loss. Setup: target elevation angle at 45 degrees, VSAT terminals, Ka-band. [Cases 1, 2, 3]

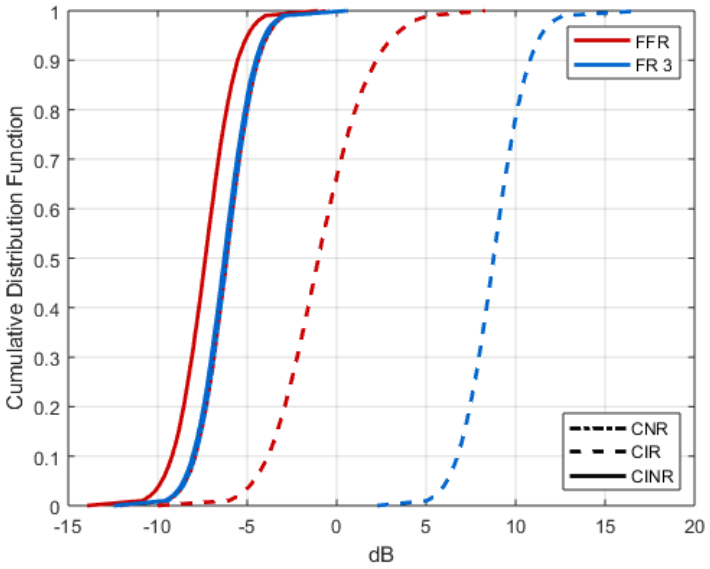

Figure 11 - GEO uplink link budget calibration. Setup: target elevation angle at 45 degrees, handheld terminals, S-band. [Cases 4, 5]

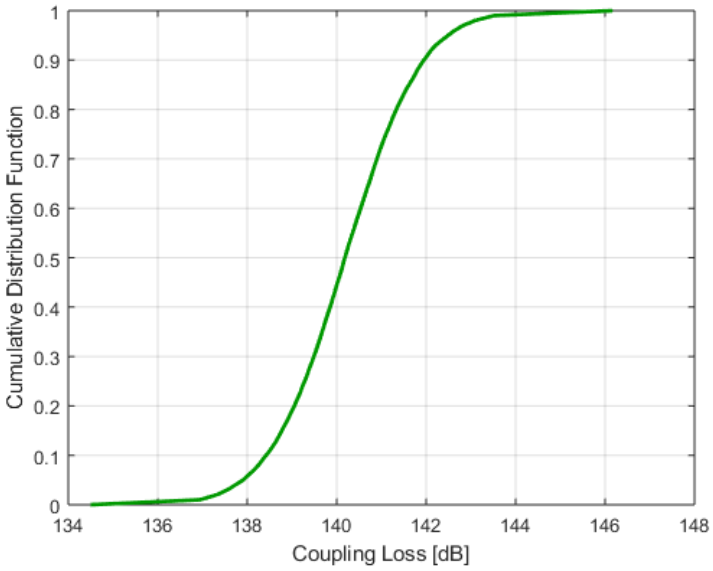

Figure 12 - GEO uplink coupling loss. Setup: target elevation angle at 45 degrees, handheld terminals, S-band. [Cases 4, 5]

3.2 LEO

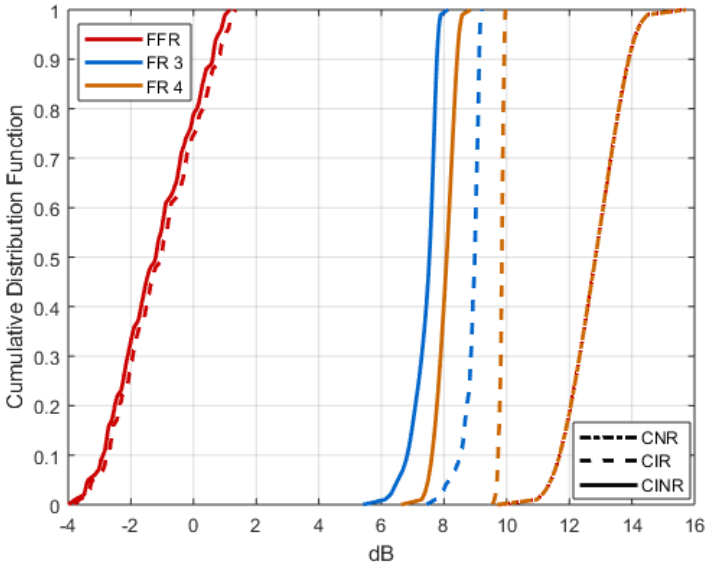

Figure 13 - LEO downlink link budget calibration. Setup: target elevation angle at 90 degrees, VSAT terminals, Ka-band, 600 km. [Cases 6, 7, 8]

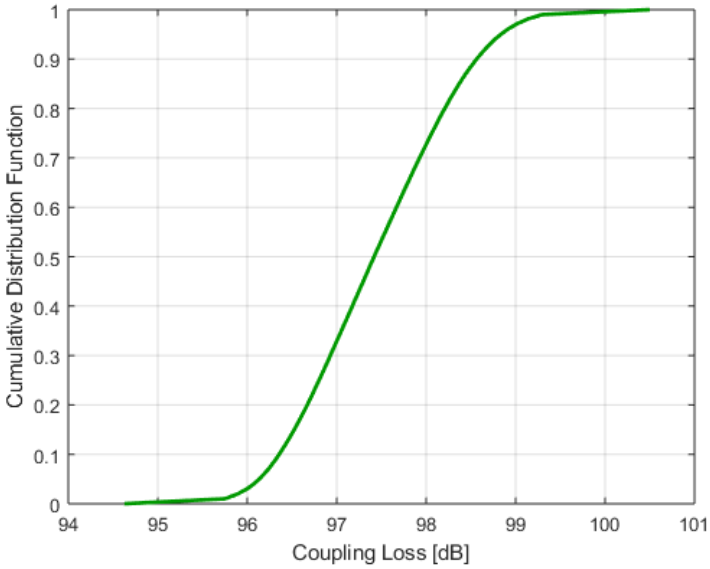

Figure 14 - LEO downlink coupling loss. Setup: target elevation angle at 90 degrees, VSAT terminals, Ka-band, 600 km. [Cases 6, 7, 8]

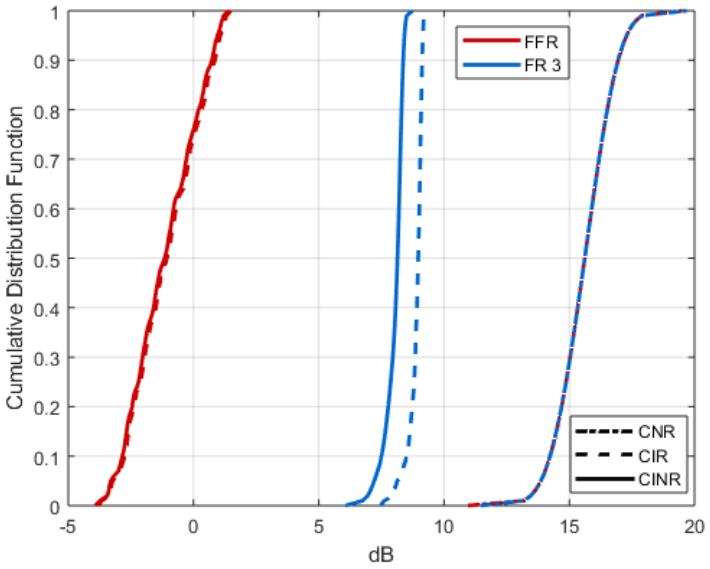

Figure 15 - LEO downlink link budget calibration. Setup: target elevation angle at 90 degrees, handheld terminals, S-band, 600 km. [Cases 9, 10]

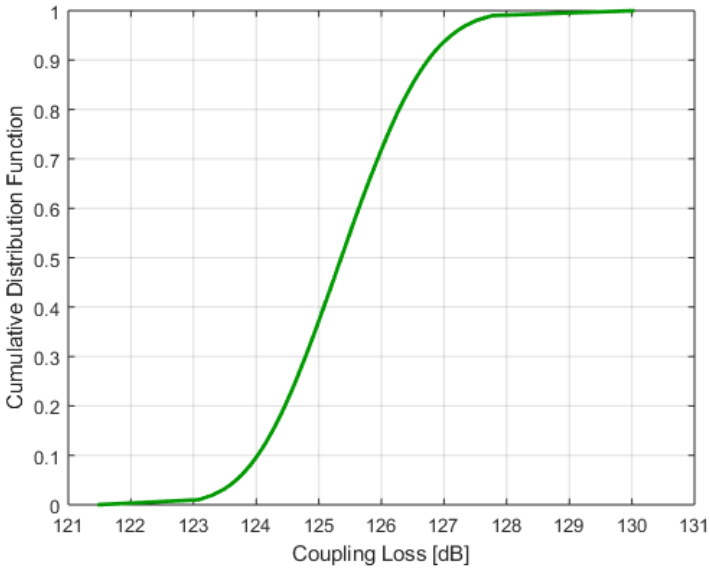

Figure 16 - LEO downlink coupling loss. Setup: target elevation angle at 90 degrees, handheld terminals, S-band, 600 km. [Cases 9, 10]

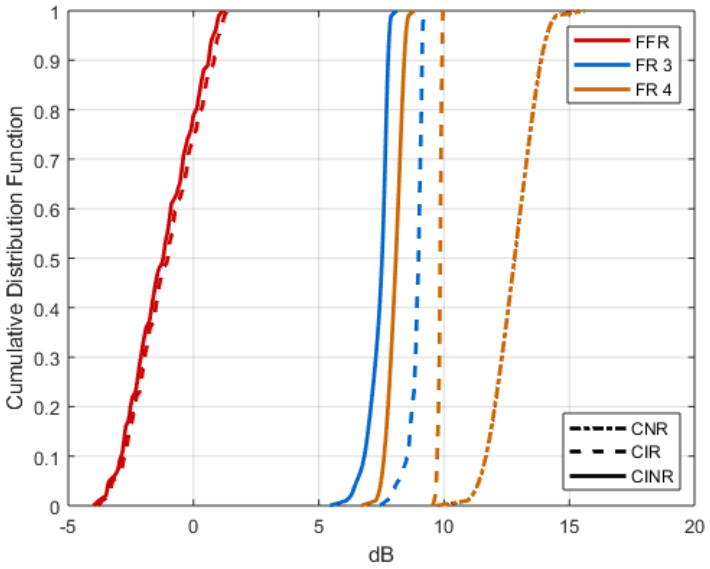

Figure 17 - LEO downlink link budget calibration. Setup: target elevation angle at 90 degrees, VSAT terminals, Ka-band, 1200 km. [Cases 11, 12, 13]

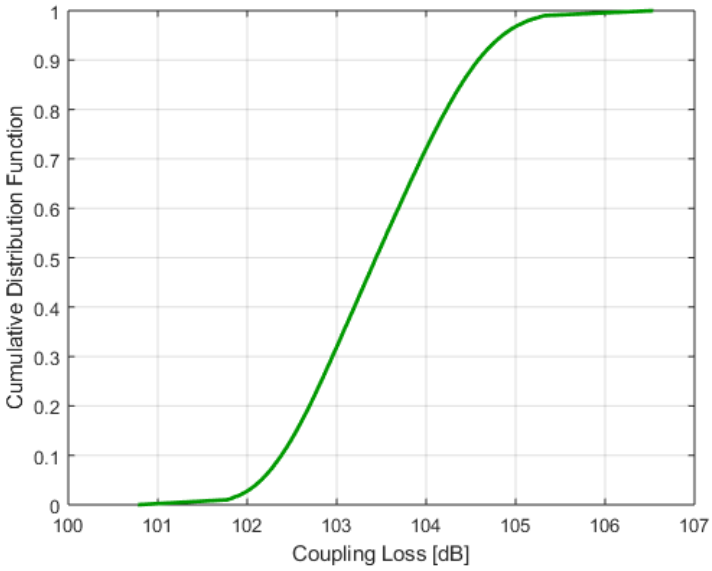

Figure 18 - LEO downlink coupling loss. Setup: target elevation angle at 90 degrees, VSAT terminals, Ka-band, 1200 km. [Cases 11, 12, 13]

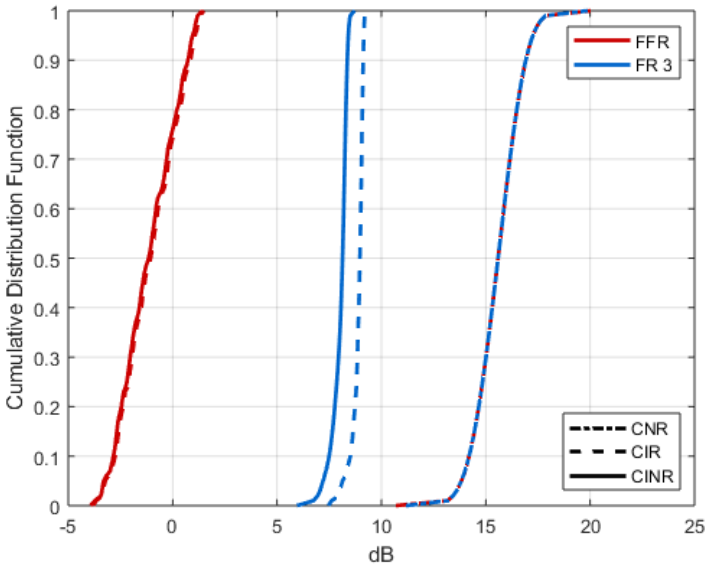

Figure 19 - LEO downlink link budget calibration. Setup: target elevation angle at 90 degrees, handheld terminals, S-band, 1200 km. [Cases 14, 15]

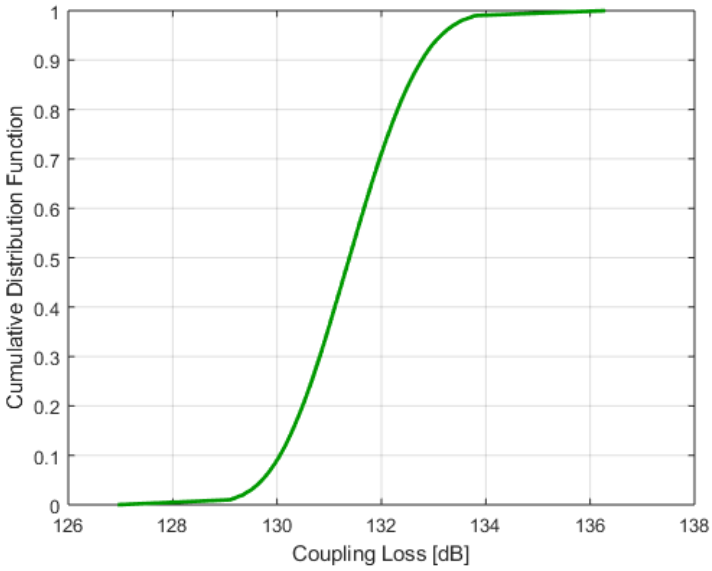

Figure 20 - LEO downlink coupling loss. Setup: target elevation angle at 90 degrees, handheld terminals, S-band, 1200 km. [Cases 14, 15]

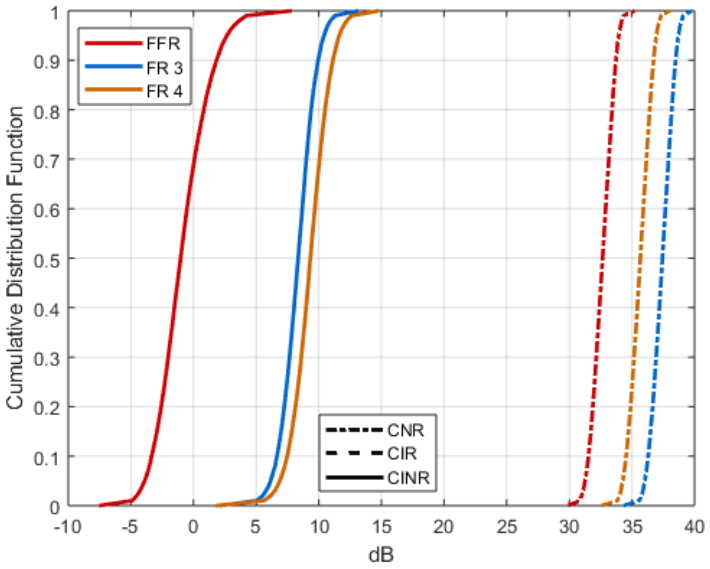

Figure 21 - LEO uplink link budget calibration. Setup: target elevation angle at 90 degrees, VSAT terminals, Ka-band, 600 km. [Cases 6, 7, 8]

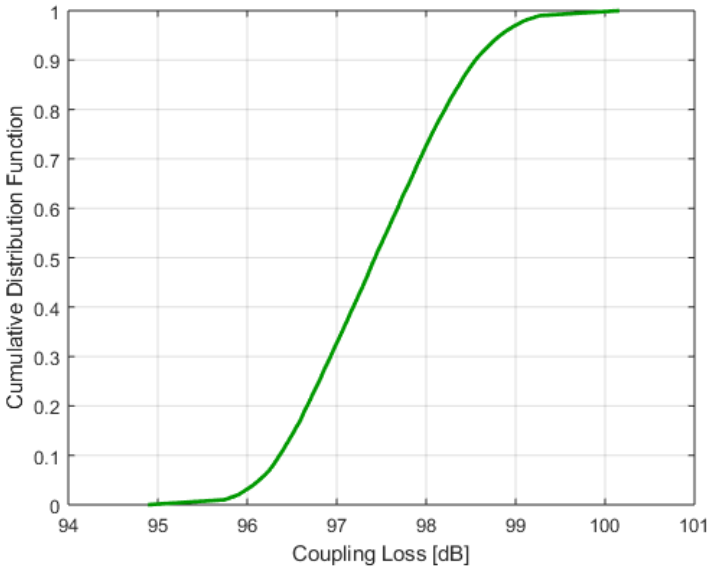

Figure 22 - LEO uplink coupling loss. Setup: target elevation angle at 90 degrees, VSAT terminals, Ka-band, 600 km. [Cases 6, 7, 8]

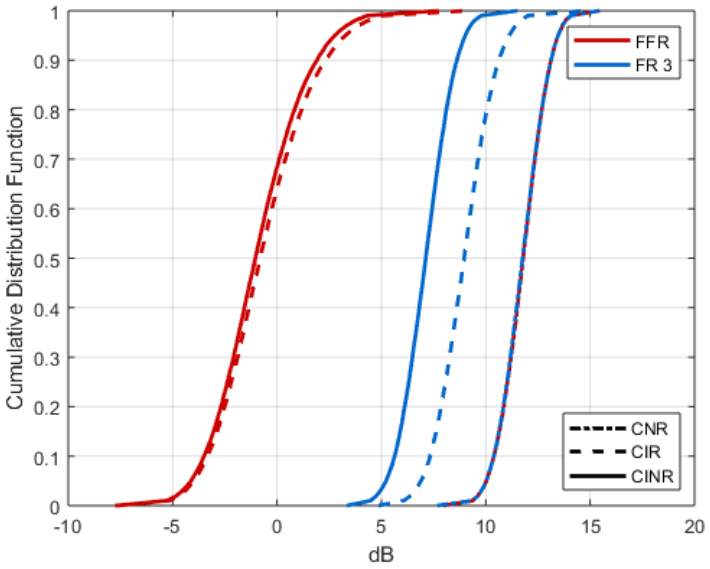

Figure 23 - LEO uplink link budget calibration. Setup: target elevation angle at 90 degrees, handheld terminals, S-band, 600 km. [Cases 9, 10]

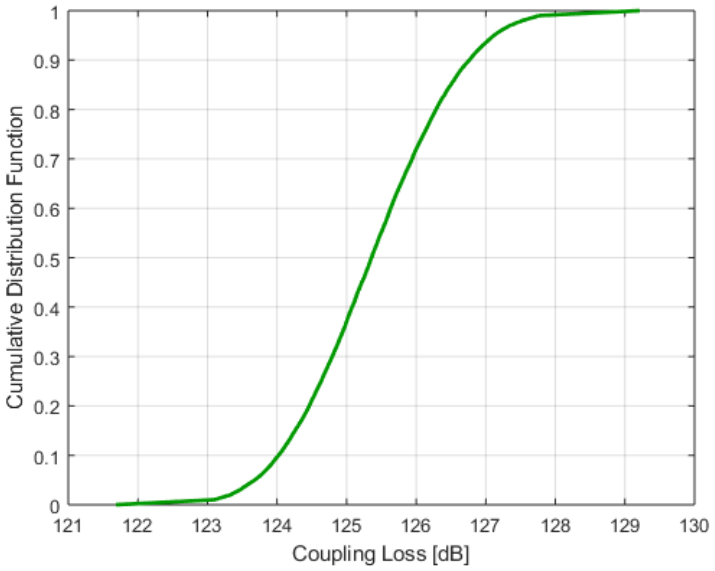

Figure 24 - LEO uplink coupling loss. Setup: target elevation angle at 90 degrees, handheld terminals, S-band, 600 km. [Cases 9, 10]

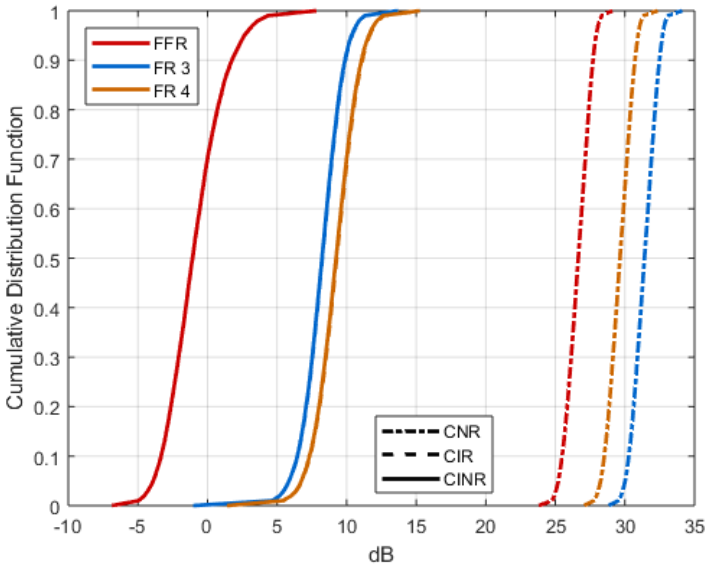

Figure 25 - LEO uplink link budget calibration. Setup: target elevation angle at 90 degrees, VSAT terminals, Ka-band, 1200 km. [Cases 11, 12, 13]

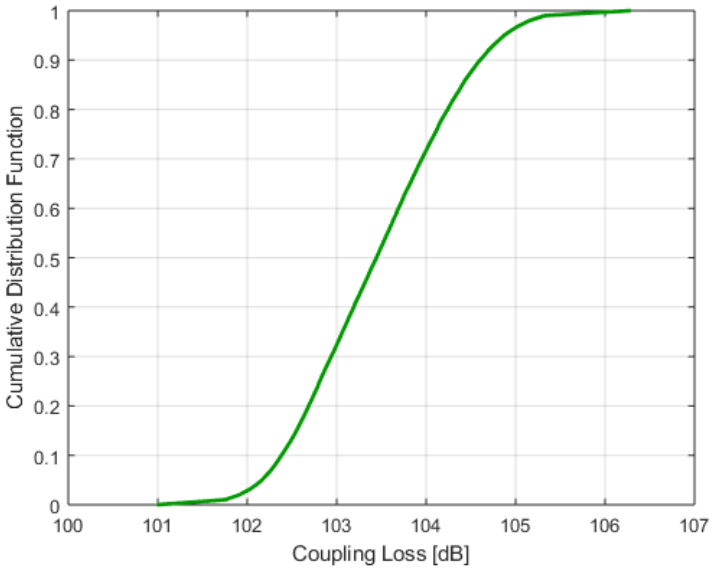

Figure 26 - LEO uplink coupling loss. Setup: target elevation angle at 90 degrees, VSAT terminals, Ka-band, 1200 km. [Cases 11, 12, 13]

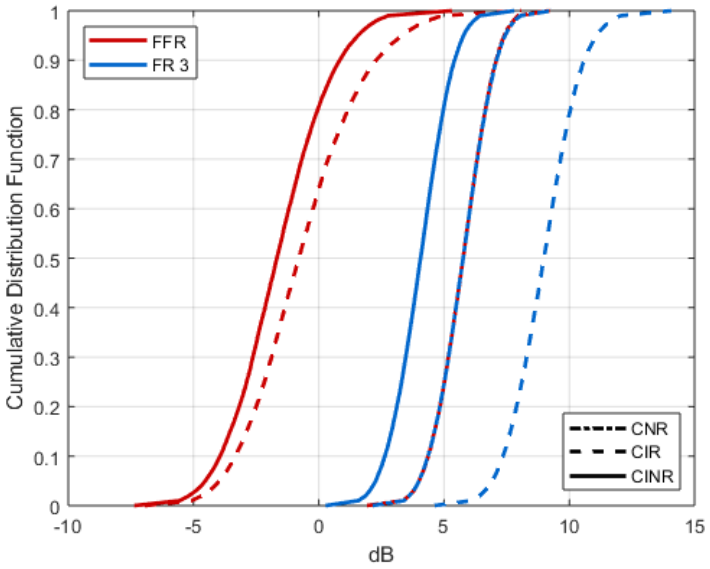

Figure 27 - LEO uplink link budget calibration. Setup: target elevation angle at 90 degrees, handheld terminals, S-band, 1200 km. [Cases 14, 15]

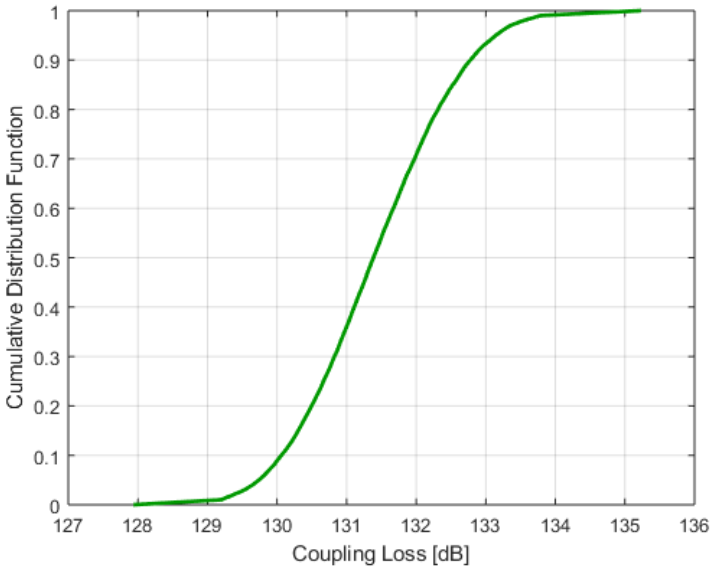

Figure 28 - LEO uplink coupling loss. Setup: target elevation angle at 90 degrees, handheld terminals, S-band, 1200 km. [Cases 14, 15]

4 Link Budget Calibration

The LB calibration methodology is reported in Sec. 6.1.3 of [3] and taking into account the RAN1 #98bis agreements [8].

For the sake of clarity, the figures below show the 19 beam-layout simulated for GEO and LEO constellation with 12.5 degrees and 30 degrees elevation angles, respectively. As agreed, the generated beams with a partial or full coverage outside the Earth are discarded.

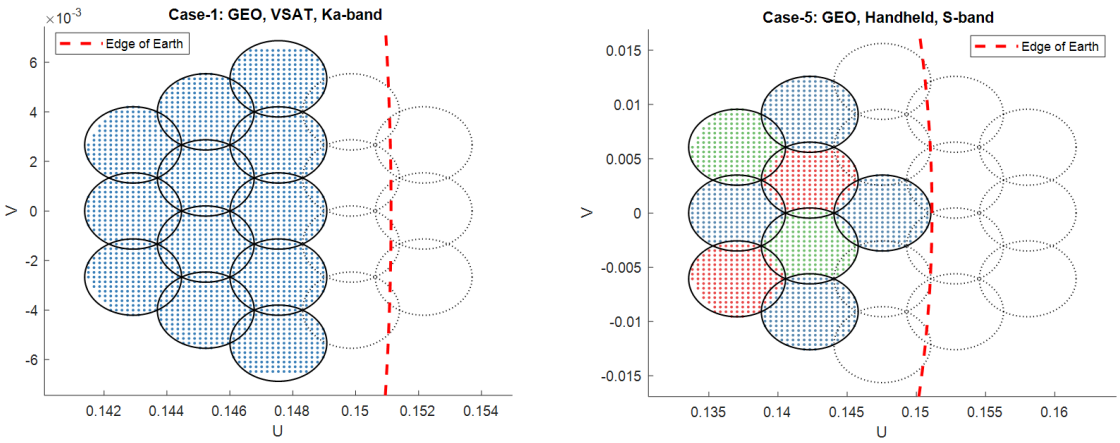

Figure 29: Beam layout for the GEO calibration case (12.5 degrees)

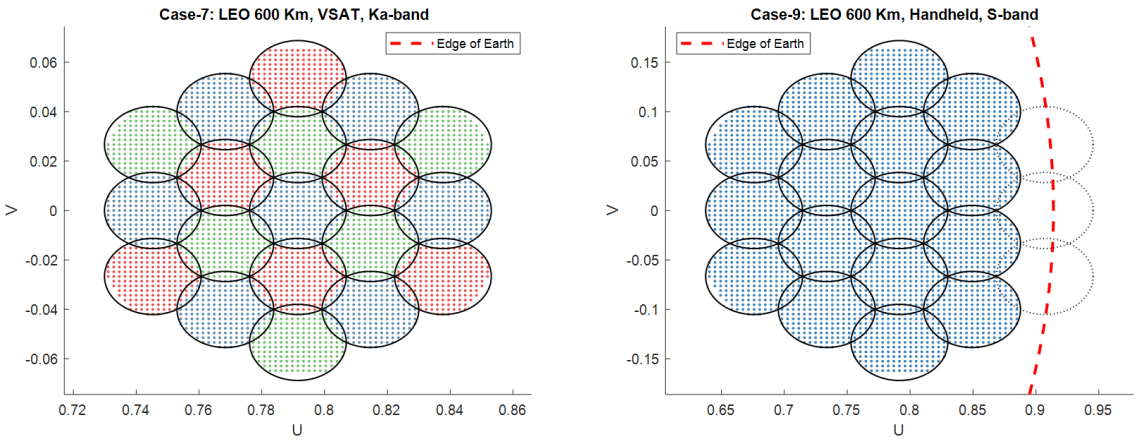

Figure 30: Beam layout for the LEO calibration case (at 600 km).

Concerning the C/I computation and, in particular, the procedure how to average the C/I values among several Monte Carlo simulations, it is not explicitly detailed in [3]. For that reason and in order to facilitate the convergence of the calibration process among Companies, we have generated the C/I statistics implementing both the linear mean and the geometric mean (*i.e.*, the mean in the dB domain).

All data values have been collected in the attached MS Excel file. Nevertheless, few examples of the obtained link budgets are summarized below.

Table 1: Link budget for the Case-1 (GEO Set-1, VSAT in Ka-band, full frequency reuse)

Companies	ESA (*geo_mean*)		ESA (*lin_mean*)	
Transmission mode	DL	UL	DL	UL
Frequency [GHz]	20.00	30.00	20.00	30.00
TX: EIRP [dBm]	96.02	76.21	96.02	76.21
RX: G/T [dB/T]	15.86	28.00	15.86	28.00
Bandwidth [MHz]	400.00	400.00	400.00	400.00
Free space path loss [dB]	210.58	214.10	210.58	214.10
Atmospheric loss [dB]	1.21	1.14	1.21	1.14
Shadow fading margin [dB]	0.00	0.00	0.00	0.00
Scintillation Loss [dB]	1.08	1.08	1.08	1.08
Polarization loss [dB]	0.00	0.00	0.00	0.00
Additional losses [dB]	0.00	0.00	0.00	0.00
CNR [dB]	11.59	0.46	11.59	0.46
CIR [dB]	1.88	1.01	2.60	1.98
CINR [dB]	1.44	-2.28	2.08	-1.86

Table 2: Link budget for the Case-4 (GEO Set-1, Handheld in S-band, full frequency reuse)

Companies	ESA (*geo_mean*)		ESA (*lin_mean*)	
Transmission mode	DL	UL	DL	UL
Frequency [GHz]	2.00	2.00	2.00	2.00
TX: EIRP [dBm]	103.77	23.01	103.77	23.01
RX: G/T [dB/T]	-31.62	19.00	-31.62	19.00
Bandwidth [MHz]	30.00	0.36	30.00	0.36
Free space path loss [dB]	190.58	190.58	190.58	190.58
Atmospheric loss [dB]	0.16	0.16	0.16	0.16
Shadow fading margin [dB]	3.00	3.00	3.00	3.00
Scintillation Loss [dB]	2.20	2.20	2.20	2.20
Polarization loss [dB]	0.00	0.00	0.00	0.00
Additional losses [dB]	0.00	0.00	0.00	0.00
CNR [dB]	0.03	-10.90	0.03	-10.90
CIR [dB]	5.88	4.47	7.80	6.06
CINR [dB]	-0.97	-11.02	-0.64	-10.98

Table 3: Link budget for the Case-10 (LEO@600 Set-1, Handheld in S-band, frequency reuse 3)

Companies	ESA (*geo_mean*)		ESA (*lin_mean*)	
Transmission mode	DL	UL	DL	UL
Frequency [GHz]	2.00	2.00	2.00	2.00
TX: EIRP [dBm]	74.00	23.01	74.00	23.01
RX: G/T [dB/T]	-31.62	1.10	-31.62	1.10
Bandwidth [MHz]	10.00	0.36	10.00	0.36
Free space path loss [dB]	159.10	159.10	159.10	159.10
Atmospheric loss [dB]	0.07	0.07	0.07	0.07
Shadow fading margin [dB]	3.00	3.00	3.00	3.00
Scintillation Loss [dB]	2.20	2.20	2.20	2.20
Polarization loss [dB]	0.00	0.00	0.00	0.00
Additional losses [dB]	0.00	0.00	0.00	0.00
CNR [dB]	6.61	2.78	6.61	2.78
CIR [dB]	12.12	12.20	12.42	13.09
CINR [dB]	5.53	2.31	5.59	2.39

5 Conclusions

This document reports a large set of system level calibration results and link budget computation for the downlink and the uplink of different satellite orbits (GEO and LEO) with the assumption of different receiver antennas. All data values have been also included in the attached MS Excel, following the agreed template.

Observation 1: Handheld have been assumed with two RX-branches, therefore no polarization loss in the downlink results.

Observation 2: For uplink results with handheld terminals, 3dB polarization loss has been considered only in case of frequency reuse 4.

Proposal 1: Consider all shown CDFs and link-budget results for the inclusion in the calibration exercise among all companies.

6 References

[1] RP-181370, "Study on solutions for NR to support non-terrestrial networks (NTN)", Thales, RAN#80, La Jolla, USA, June 11-14, 2018.

[2] RP-182444, "Study on solutions for NR to support non-terrestrial networks (NTN)", Thales, RAN#82, Sorrento, Italy, December 10-13, 2018.

[3] TR 38.821 V0.8.0 (2019-09), "Solutions on NR to support non-terrestrial networks", 3GPP source.

[4] A. Guidotti, A. Vanelli-Coralli (University of Bologna), Report on "NTN Link Budget Analysis", ESA contract, "*5G4Space*: Spin-In of 3GPP Terrestrial Radio Access Technology for SatCom", Aug. 2019.

[5] A. Guidotti, A. Vanelli-Coralli, S. Cioni, A. Mengali, "Non-Terrestrial Networks: Link Budget Analysis", submitted to IEEE ICC 2020, Sep. 2019.

[6] R1-1911254, "NTN link budget calibration: downlink numerical results", RAN1 #98bis, Oct. 2019.

[7] TR 38.811 V15.1.0 (2019-06), "Study on NR to support non terrestrial networks", 3GPP source.

[8] RAN1 Chairman's notes, RAN1 #98bis, Oct. 2019.